

Introduction to CNS Pharmacology

TARGETS OF CNS DRUG ACTION

Drugs that act on the central nervous system (CNS) are the most commonly prescribed drugs in current use. Most of these drugs act by changing ion flow through transmembrane channels of nerve cells. Transmitter reuptake transporters constitute a second class of drug targets, especially for antidepressant agents. Inhibition of acetylcholine metabolism is the major action of the drugs currently approved for use in Alzheimer disease and γ -aminobutyric acid (GABA) metabolism is inhibited by an anticonvulsant agent. Finally, a few drugs appear to act by altering the function of neuroglia. These satellite cells have been shown to modulate transmitter synthesis and disposition and support neurons metabolically. Microglia have also been shown to “prune” neuronal networks in the normal development of the CNS and possibly in Alzheimer disease and schizophrenia.

A. Types of Ion Channels

Ion channels of neuronal membranes are of two major types: **voltage gated** and **ligand gated** (Figure 21-1). Voltage-gated ion channels respond to changes in membrane potential. They are found in high concentration on the axons of nerve cells and include the sodium channels responsible for action potential propagation. Cell bodies, axon terminals, and dendrites have voltage-sensitive ion channels for sodium, potassium, and calcium. Ligand-gated ion channels, also called ionotropic receptors, respond to chemical neurotransmitters that bind to receptor subunits present in their macromolecular structure. Neurotransmitters also bind to G-protein-coupled receptors (metabotropic receptors) that can modulate voltage-gated ion channels. Neurotransmitter-coupled ion channels are found on cell bodies and on both the presynaptic and postsynaptic sides of synapses.

B. Types of Receptor-Channel Coupling

In the case of ligand-gated ion channels, activation (or inhibition) is initiated by the interaction between chemical neurotransmitters and their receptors (Figure 21-1). Coupling may be through a receptor that acts directly on the channel protein (panel B), through a receptor that is coupled to the ion channel through a G protein (panel C), or through a receptor coupled to a G protein that modulates the formation of diffusible second messengers, including cyclic adenosine monophosphate (cAMP), inositol triphosphate (IP₃), and diacylglycerol (DAG), which secondarily modulate ion channels (panel D).

C. Role of the Ion Current Carried by the Channel

Excitatory (depolarizing) postsynaptic potentials (EPSPs) are usually generated by the opening of sodium or calcium channels. In some synapses, similar depolarizing potentials result from the *closing* of potassium channels. Inhibitory (hyperpolarizing) postsynaptic potentials (IPSPs) are usually generated by the opening of potassium or chloride channels. For example, activation of postsynaptic metabotropic receptors increases the efflux of potassium. Presynaptic inhibition can occur via a decrease in calcium influx elicited by activation of metabotropic receptors.

SITES & MECHANISMS OF DRUG ACTION

A small number of neuropharmacologic agents exert their effect through direct interactions with molecular components of channels on axons. Examples include certain anticonvulsants

High-Yield Terms to Learn

Voltage-gated ion channels
Ligand-gated ion channels
Metabotropic receptors
EPSP
IPSP
Synaptic mimicry

Transmembrane ion channels regulated by changes in membrane potential

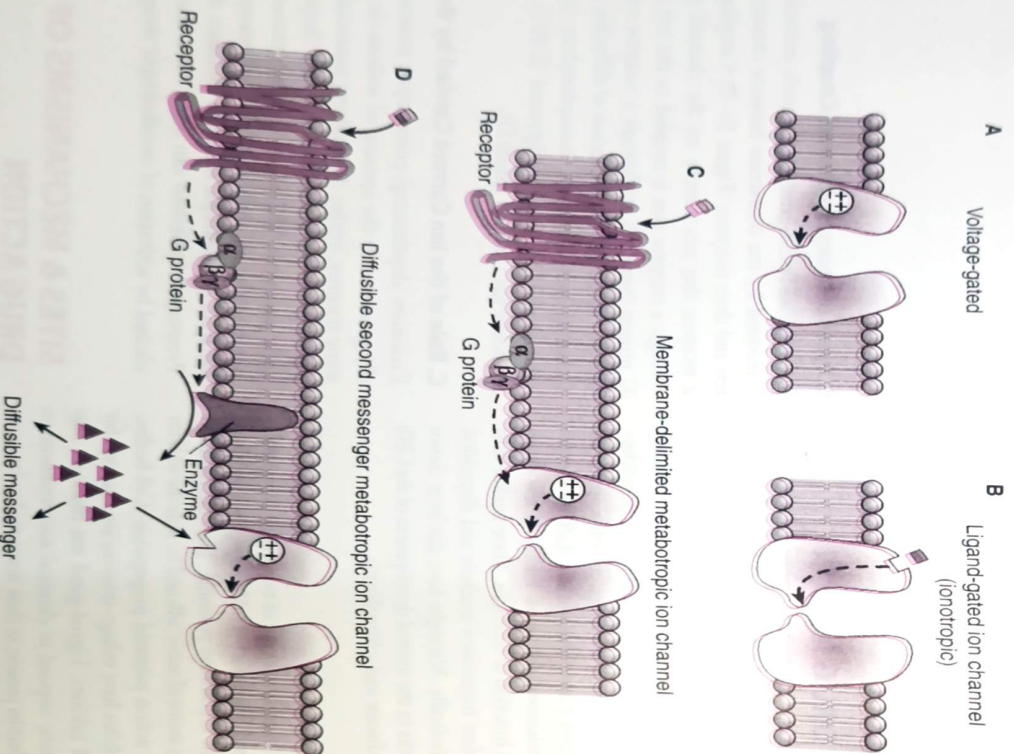
Transmembrane ion channels that are regulated by interactions between neurotransmitters and their receptors (also called ionotropic receptors)

G-protein-coupled receptors that respond to neurotransmitters either by a direct action of G proteins on ion channels or by G protein-enzyme activation that leads to formation of diffusible second messengers and change channel function

Excitatory postsynaptic potential; a depolarizing membrane potential change

Inhibitory postsynaptic potential; a hyperpolarizing membrane potential change

Ability of an administered chemical to mimic the actions of the natural neurotransmitter; a criterion for identification of a putative neurotransmitter



JRE 21-1 Types of ion channels and neurotransmitter receptors in the CNS. **A** shows a voltage-gated ion channel in which the voltage controls the gating (broken arrow). **B** shows a ligand-gated ion channel in which binding of the neurotransmitter to the ionotropic channel controls the gating. **C** shows a metabotropic receptor coupled to a G protein that can interact directly with an ion channel. **D** shows a receptor coupled to a G protein that activates an enzyme; the activated enzyme generates a diffusible second messenger, for example, cAMP, which interacts to modulate an ion channel. (Reproduced with permission from Katzung BG, Vanderah TW: *Basic & Clinical Pharmacology*, New York, NY: McGraw Hill; 2021.)

The CNS contains two types of neuronal systems: hierarchical and diffuse.

A. Hierarchical Systems

These systems are delimited in their anatomic distribution and generally contain large myelinated, rapidly conducting fibers. Hierarchical systems control major sensory and motor functions. The major excitatory transmitters in these systems are aspartate and glutamate. These systems also include numerous small inhibitory interneurons, which use GABA or glycine as transmitters. Drugs that affect hierarchical systems often have profound effects on the overall excitability of the CNS.

B. Diffuse Systems

Diffuse or nonspecific systems are broadly distributed, with single cells frequently sending branches to many different areas. The axons are fine and branch repeatedly to form synapses with many cells. Axons commonly have periodic enlargements (varicosities) that contain transmitter vesicles. The transmitters in diffuse systems are often amines (norepinephrine, dopamine, serotonin) or peptides that commonly exert actions on metabotropic receptors. Drugs that affect these systems often have marked effects on such CNS functions as attention, appetite, and emotional states.

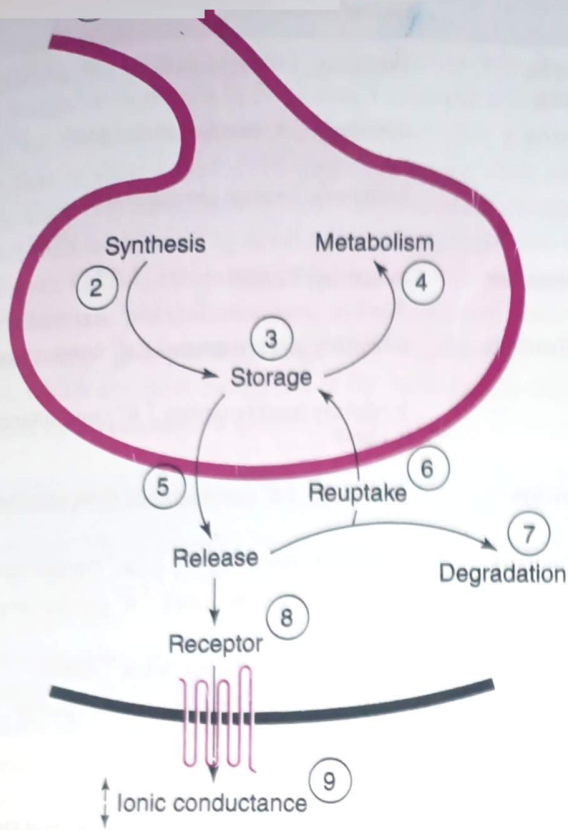


FIGURE 21-2 Sites of CNS drug action. Drugs may alter (1) the action potential in the presynaptic fiber; (2) synthesis of transmitter; (3) storage; (4) metabolism; (5) release; (6) reuptake; (7) degradation; (8) receptor for the transmitter; or (9) receptor-induced decrease or increase in ionic conductance. (Modified with permission from Katzung BG, Vanderah TW: *Basic & Clinical Pharmacology*, 15th ed. New York, NY: McGraw Hill; 2021.)

(eg, carbamazepine, phenytoin), local anesthetics, and some drugs used in general anesthesia. However, the effects of most therapeutically important CNS drugs are exerted mainly at synapses. Possible mechanisms are indicated in Figure 21-2. Thus, as noted earlier, drugs may act presynaptically to alter the synthesis, storage, release, reuptake, or metabolism of transmitter chemicals. Other drugs can activate or block both pre- and postsynaptic receptors for specific transmitters or can interfere with the actions of second messengers. The selectivity of CNS drug action is largely based on the fact that different groups of neurons use different neurotransmitters and that they are segregated into networks that subserve different CNS functions.

Drugs that alter transmitter reuptake (site 6 in Figure 21-2) are specially important for 5-HT, noradrenergic, and dopaminergic synapses. Serotonin and norepinephrine transporters (SERT and NET, respectively) are the targets of antidepressants, while NET and the dopamine reuptake transporter are the targets of amphetamines and cocaine.

A few neurotoxic substances damage or kill nerve cells. For example, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) is cytotoxic to neurons of the nigrostriatal dopaminergic pathway.

TRANSMITTERS AT CENTRAL SYNAPSES

A. Criteria for Transmitter Status

To be accepted as a neurotransmitter, a candidate chemical must (1) be present in higher concentration in the synaptic area than in other areas (ie, must be localized in appropriate areas), (2) be released by electrical or chemical stimulation via a calcium-dependent mechanism, and (3) produce the same sort of postsynaptic response that is seen with physiologic activation of the synapse (ie, must exhibit synaptic mimicry). Table 21-1 lists the most important chemicals currently accepted as neurotransmitters in the CNS.

B. Glutamic Acid

Glutamic acid is the **most important excitatory transmitter** in the CNS: most neurons in the brain are excited by glutamic acid. High concentrations of glutamic acid in synaptic vesicles are achieved by the vesicular glutamate transporter (VGLUT). Both ionotropic and metabotropic receptors have been characterized. There are three major subtypes of glutamate-responsive channels. The channel associated with the *N*-methyl-D-aspartate (NMDA) receptor mediates increased Na^+ , K^+ , and Ca^{2+} current. The NMDA receptor is blocked by phencyclidine (PCP) and ketamine and its channel conductance is correspondingly decreased. NMDA receptors appear to play a role in synaptic plasticity related to learning and memory via the **long-term potentiation** (LTP) phenomenon. **Memantine** is an NMDA antagonist approved for treatment of Alzheimer dementia. Excessive activation of NMDA

TABLE 21-1 Major neurotransmitters in the CNS.

Transmitter	Anatomical Distribution	Receptor Subtypes	Receptor Mechanisms
Acetylcholine	Cell bodies at all levels, short and long axons	Muscarinic, M ₁ ; blocked by pirenzepine and atropine	Excitatory; ↓ K ⁺ conductance; ↑ IP ₃ and DAG
		Muscarinic, M ₂ ; blocked by atropine	Inhibitory; ↑ K ⁺ conductance; cAMP
	Motoneuron-Renshaw cell synapse	Nicotinic, N	Excitatory; ↑ cation conductance
Dopamine	Cell bodies at all levels, short, medium, and long axons	D ₁ ; blocked by phenothiazines	Inhibitory; ↑ cAMP
		D ₂ ; blocked by phenothiazines and haloperidol	Inhibitory (presynaptic); ↓ Ca ²⁺ conductance
			Inhibitory (postsynaptic); ↑ K ⁺ conductance; ↓ cAMP
Norepinephrine	Cell bodies in pons and brain stem project to all levels	Alpha ₁ ; blocked by prazosin	Excitatory; ↓ K ⁺ conductance; ↑ IP ₃ and DAG
		Alpha ₂ ; activated by clonidine	Inhibitory (presynaptic); ↓ Ca ²⁺ conductance
			Inhibitory (postsynaptic); ↑ K ⁺ conductance; ↓ cAMP
		Beta ₁ ; blocked by propranolol Beta ₂ ; blocked by propranolol	Excitatory; ↓ K ⁺ conductance; ↑ cAMP Inhibitory; ↑ electrogenic sodium pump
Serotonin (5-hydroxytryptamine)	Cell bodies in midbrain and pons project to all levels	5-HT _{1A} ; buspirone is a partial agonist	Inhibitory; ↑ K ⁺ conductance
		5-HT _{2A} ; blocked by clozapine, risperidone, and olanzapine	Excitatory; ↓ K ⁺ conductance; ↑ IP ₃ and DAG
		5-HT ₃ ; blocked by ondansetron	Excitatory; ↑ cation conductance
		5-HT ₄	Excitatory; ↓ K ⁺ conductance; ↑ cAMP
GABA	Supraspinal interneurons; spinal interneurons involved in presynaptic inhibition	GABA _A ; facilitated by benzodiazepines and zolpidem	Inhibitory; ↑ Cl ⁻ conductance
		GABA _B ; activated by baclofen	Inhibitory (presynaptic); ↓ Ca ²⁺ conductance Inhibitory (postsynaptic); ↑ K ⁺ conductance
Glutamate, aspartate	Relay neurons at all levels	Four subtypes; NMDA subtype blocked by phencyclidine, ketamine, and memantine	Excitatory; ↑ Ca ²⁺ or cation conductance
		Metabotropic subtypes	Inhibitory (presynaptic); ↓ Ca ²⁺ conductance; ↓ cAMP
			Excitatory (postsynaptic); ↓ K ⁺ conductance; ↑ IP ₃ and DAG
Glycine	Interneurons in spinal cord and brain stem	Single subtype; blocked by strychnine	Inhibitory; ↑ Cl ⁻ conductance
Opioid peptides	Cell bodies at all levels	Three major subtypes: μ, δ, κ	Inhibitory (presynaptic); ↓ Ca ²⁺ conductance; ↓ cAMP
			Inhibitory (postsynaptic); ↑ K ⁺ conductance; ↓ cAMP

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receptors after neuronal injury may be responsible for cell death. The second glutamate receptor type, the **AMPA** receptors, mediate an excitatory postsynaptic increase in Na⁺ and K⁺ but not Ca²⁺ conductance. The third glutamate receptor type, **kainate** receptors, also activate a Na⁺-K⁺ channel. An important marine toxin,

domoic acid, activates the kainate receptors and causes excitatory toxicity. Activation of several **glutamate metabotropic receptors** can reduce presynaptic Ca²⁺ conductance, decrease cAMP, and decrease K⁺ conductance; and postsynaptically increase IP₃ and DAG, resulting in increased intracellular Ca²⁺ release.

C. GABA and Glycine

GABA is the **primary inhibitory neurotransmitter** mediating IPSPs in neurons in the brain; it is also important in the spinal cord. **GABA_A** receptor activation opens chloride ion channels. **GABA_B** receptors (activated by **baclofen**, a centrally acting muscle relaxant) are coupled to G proteins that either open a potassium channel that is shared with 5-HT_{1A} receptors, or close calcium channels. Fast IPSPs are blocked by GABA_A receptor antagonists, and slow IPSPs are blocked by GABA_B receptor antagonists. Drugs that influence GABA_A receptor systems include sedative-hypnotics (eg, **barbiturates**, **benzodiazepines**, **zolpidem**) and some anti-convulsants (eg, **gabapentin**, **tiagabine**, **vigabatrin**). Glycine receptors, which are more numerous in the cord than in the brain, are blocked by **strychnine**, a spinal convulsant.

D. Acetylcholine

Approximately 5% of brain neurons have receptors for acetylcholine (ACh). Most CNS responses to ACh are mediated by a large family of G-protein-coupled muscarinic M₁ receptors that lead to slow excitation when activated. The ionic mechanism of slow excitation involves a *decrease* in membrane potassium permeability. Of the nicotinic receptors present in the CNS (they are less common than muscarinic receptors), those on the Renshaw cells activated by motor axon collaterals in the spinal cord are the best characterized. Drugs affecting the activity of cholinergic systems in the brain include the acetylcholinesterase inhibitors used in Alzheimer disease (eg, **rivastigmine**) and the muscarinic blocking agents used in parkinsonism (eg, **benztropine**).

E. Dopamine

Dopamine exerts slow inhibitory actions at synapses in specific neuronal systems, commonly via G-protein-coupled activation of potassium channels (postsynaptic) or inhibition of calcium channels (presynaptic). The D₂ receptor is the main dopamine subtype in basal ganglia neurons, and it is widely distributed at the supraspinal level. Dopaminergic pathways include the nigro-striatal, mesolimbic, and tuberoinfundibular tracts. In addition to the two receptors listed in Table 21-1, three other dopamine receptor subtypes have been identified (D₃, D₄, and D₅). Drugs that block the activity of dopaminergic pathways include older antipsychotics (eg, **chlorpromazine**, **haloperidol**), which may cause parkinsonian symptoms. Drugs that increase synaptic dopaminergic activity include CNS stimulants (eg, **amphetamine**), and commonly used antiparkinsonism drugs (eg, **levodopa**).

F. Norepinephrine

Noradrenergic neuron cell bodies are mainly located in the brain stem and the lateral tegmental area of the pons. These neurons fan out broadly to provide most regions of the CNS with diffuse noradrenergic input. Excitatory effects are produced by activation of α_1 and β_1 receptors. Inhibitory effects are caused by activation of α_2 and β_2 receptors. Norepinephrine receptors appear to be involved in mood, appetite, and alertness. CNS stimulants (eg, **amphetamines** [used in attention deficit hyperactivity disorder, ADHD], **cocaine**), **monoamine oxidase inhibitors** (eg, **phenelzine**), and

tricyclic antidepressants (eg, **amitriptyline**) are examples of drugs that enhance the activity of noradrenergic pathways.

G. Serotonin

Most serotonin (5-hydroxytryptamine; 5-HT) pathways originate from cell bodies in the raphe or midline regions of the pons and upper brain stem; these pathways innervate most regions of the CNS. Multiple 5-HT receptor subtypes have been identified and, with the exception of the 5-HT₂ subtype, all are metabotropic. 5-HT_{1A} receptors and GABA_B receptors share the same potassium channel. Serotonin can cause excitation or inhibition of CNS neurons depending on the receptor subtype activated. Both excitatory and inhibitory actions can occur on the same neuron if appropriate receptors are present. Most of the agents used in the treatment of major depressive disorders affect serotonergic pathways to some degree (eg, tricyclic antidepressants, selective serotonin reuptake inhibitors, serotonin-norepinephrine reuptake inhibitors). The actions of some CNS stimulants and newer antipsychotic drugs (eg, **olanzapine**) also appear to be mediated via effects on serotonergic transmission. **Reserpine**, which may cause severe depression of mood, depletes vesicular stores of both serotonin and norepinephrine in CNS neurons.

H. Peptide Transmitters

Many peptides have been identified in the CNS, and some meet most or all of the criteria for acceptance as neurotransmitters. The best-defined peptides are the **opioid peptides** (**β -endorphin**, **met-enkephalin**, and **dynorphin**), which are distributed at all levels of the neuraxis. Some of the important therapeutic actions of opioid analgesics (eg, **morphine**) are mediated via activation of receptors for these endogenous peptides. Another peptide, **substance P**, is a mediator of slow EPSPs in neurons involved in nociceptive sensory pathways in the spinal cord and brain stem. **Orexins** are peptides associated with the sleep-wake cycle and promote wakefulness. Peptide transmitters differ from nonpeptide transmitters in that (1) the peptides are synthesized in the cell body and transported to the nerve ending via axonal transport, and (2) no reuptake or specific enzyme mechanisms have been identified for terminating their actions.

I. Endocannabinoids

These are widely distributed brain lipid derivatives (eg, 2-arachidonylglycerol) that bind to receptors for the cannabinoids found in marijuana (see Chapter 60). They are synthesized and released postsynaptically after membrane depolarization but diffuse backward (retrograde), acting presynaptically to decrease transmitter release, via their interaction with a specific cannabinoid receptor.

J. Other Transmitters

Histamine receptors are widely distributed in the brain and appear to modulate arousal, appetite, and memory. Centrally acting antihistamines have significant sedative and antimotion sickness effects (see Chapter 16). **Nitric oxide (NO)** is not stored but synthesized on demand. Nitric oxide synthase is found in several types of neurons but the role of NO in CNS function is poorl

understood. **Purines** are cotransmitters in several types of transmitter vesicles and receptors for ATP, adenosine, UTP, and UDP are found in the CNS. ATP acts on metabotropic A_1 receptors to inhibit calcium channels and reduce the release of other transmitters. ATP also acts on P2X and P2Y receptors.

QUESTIONS

- An increase of chloride conductance is the most important result of activation of
 - Acetylcholine M_1 receptors
 - Beta $_1$ adrenoceptors
 - 5-HT $_3$ receptors
 - GABA $_A$ receptors
 - Glutamate NMDA receptors
- Neurotransmitters may
 - Increase chloride conductance to cause hyperpolarization
 - Increase potassium conductance to cause inhibition
 - Increase sodium conductance to cause excitation
 - Inhibit calcium conductance to reduce transmitter release
 - Exert all of the above actions
- All of the listed neurotransmitters change membrane excitability by decreasing K^+ conductance *except*
 - Acetylcholine
 - Dopamine
 - Glutamic acid
 - Norepinephrine
 - Serotonin
- Which of the following receptors shares the same potassium channel as the 5-HT $_{1A}$ receptor?
 - Dopamine D_2 receptor
 - GABA $_B$ receptor
 - Mu opioid receptor
 - Muscarinic M_1 receptor
 - Substance P receptor
- Which of the following chemicals is most likely to function as a neurotransmitter in hierarchical system-2?
 - Glutamate
 - Met-enkephalin
- This transmitter is found in the CNS. It appears to be involved in states, and its function is to
 - Acetylcholine
 - Dopamine
 - GABA
 - Glutamate
 - Serotonin
- Which of the following is a neurotransmitter for a neuromuscular junction?
 - Acetylcholine
 - Cyclo-oxygenase
 - Dopamine
 - Glycine
 - Sulfate
- One of the following is a neurotransmitter in the CNS. Activation of this receptor leads to an increase in cation conductance coupled with a decrease in potassium conductance. Which of the following is most likely to be the neurotransmitter?
 - Dopamine
 - Glutamate
 - GABA
 - Norepinephrine
 - Serotonin
- This neurotransmitter is found in both excitatory and inhibitory types of synapses. It is targeted to both presynaptic and postsynaptic receptors. Which of the following is most likely to be the neurotransmitter?
 - Acetylcholine
 - Beta-endorphin
 - GABA
 - Glutamate
 - Norepinephrine